VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date: 26/02/2026**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: TARUN KUMAR **Reg. no.: 23BCE7563**

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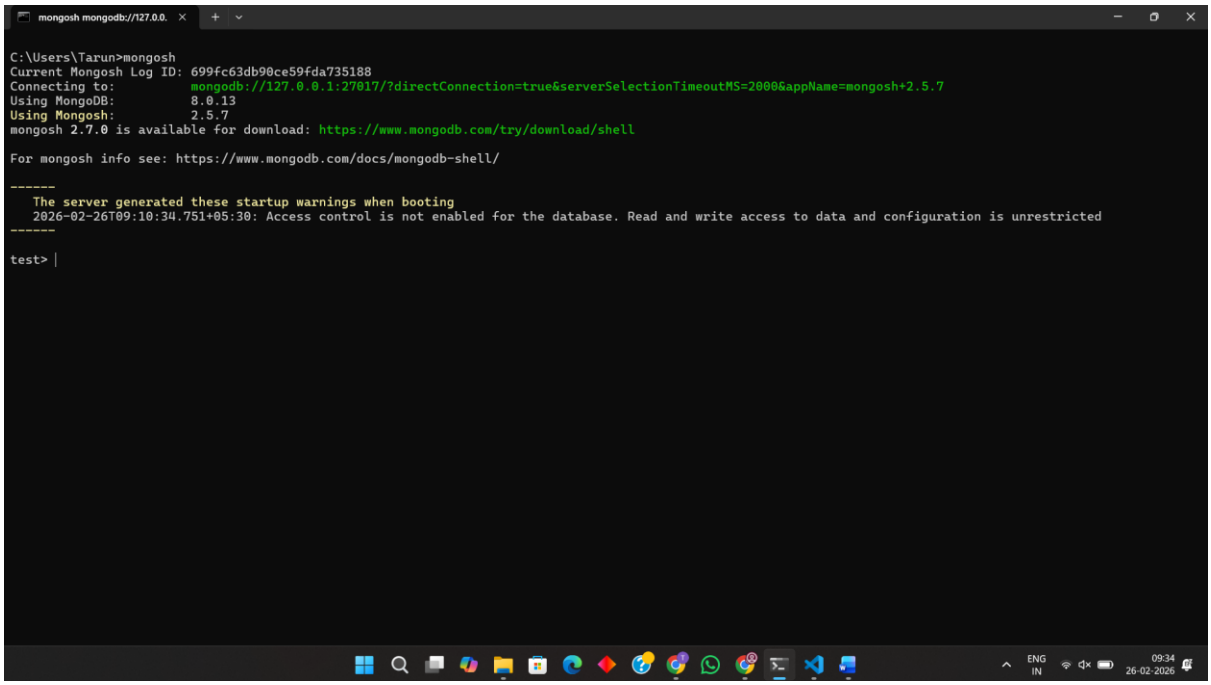
1. Use MongoDB to implement the following DB operations



1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".

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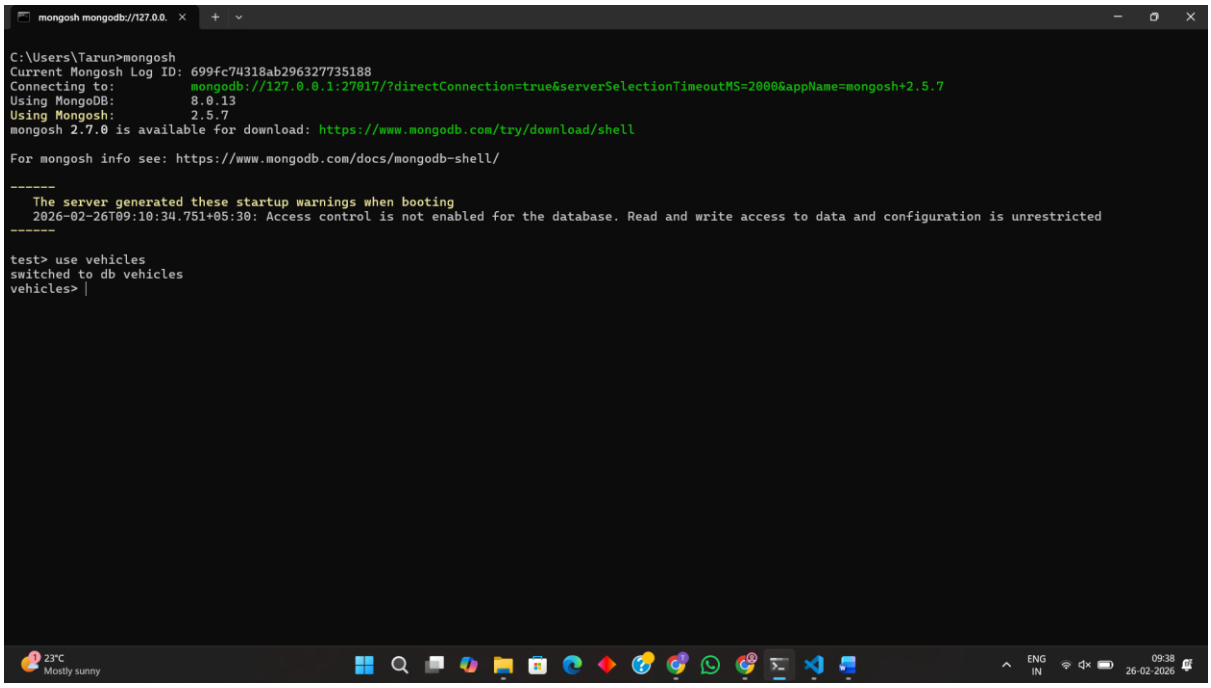
use vehicles



2. Write a MongoDB query to display all the databases.

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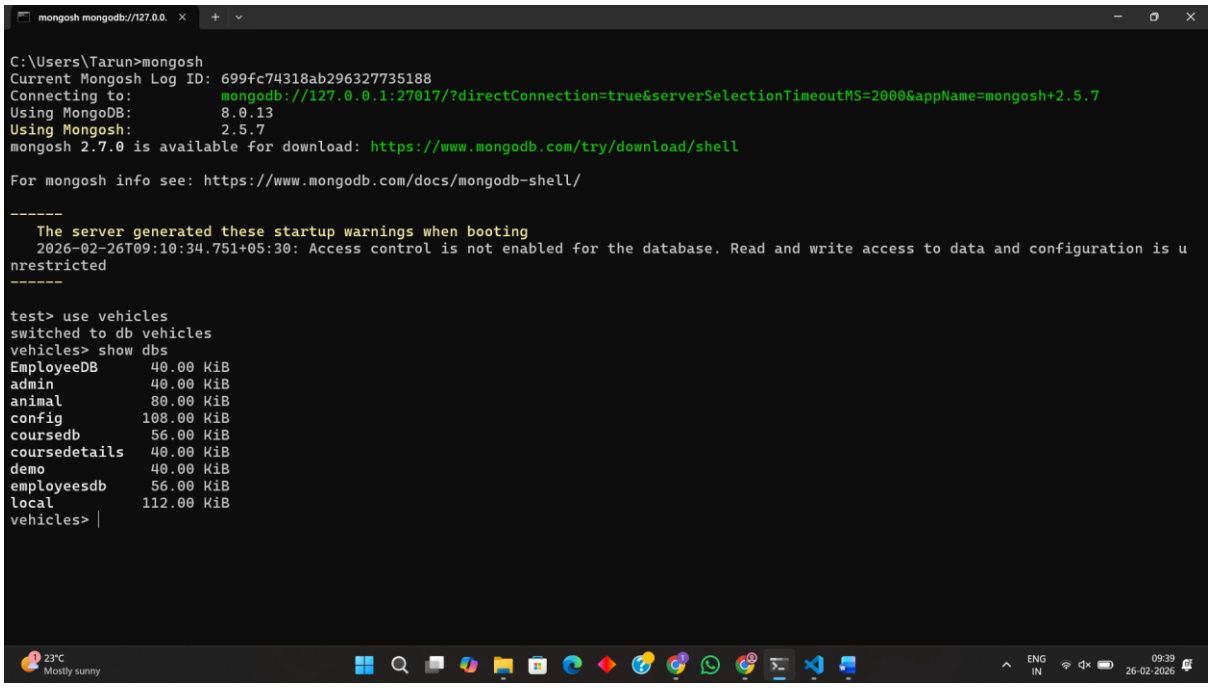
show dbs



3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.

db.createCollection("two_wheelers", { capped: true, size: 4194304, max: 2000 })
db.createCollection("four_wheelers")

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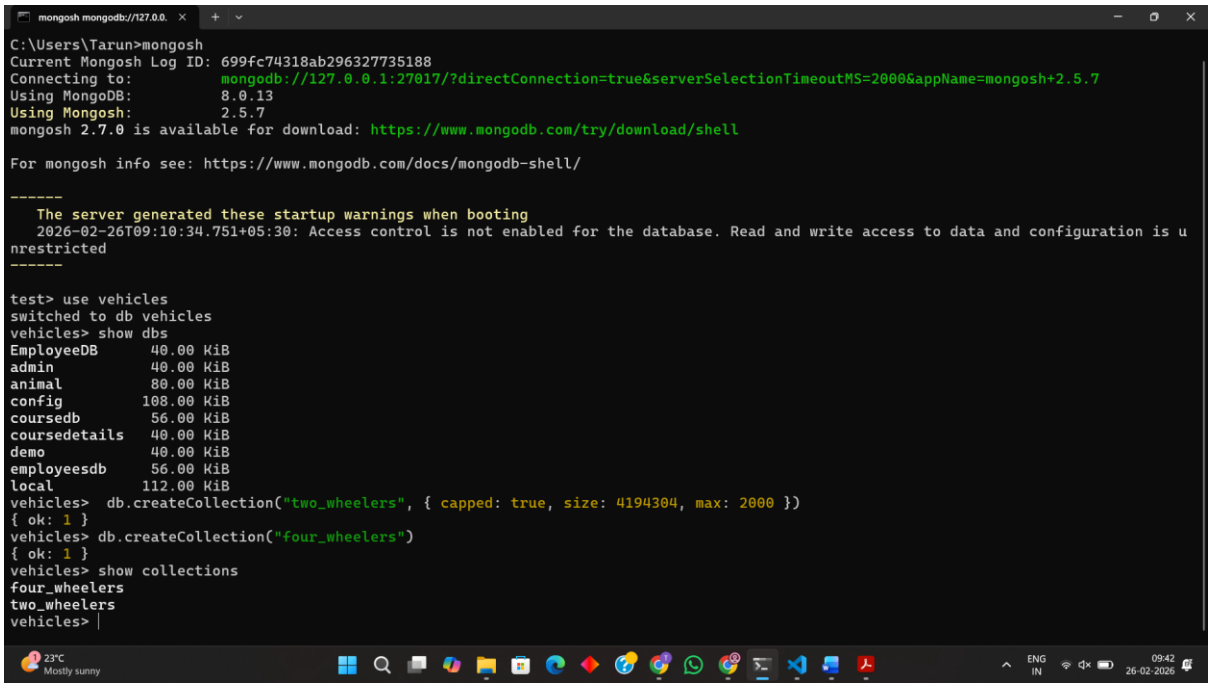

```
C:\Users\Tarun>mongosh
Current Mongosh Log ID: 699fc74318ab296327735188
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.7
Using MongoDB:          8.0.13
Using Mongosh:          2.5.7
mongosh 2.7.0 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-02-26T09:10:34.751+05:30: Access control is not enabled for the database. Read and write access to data and configuration is u
nrestricted
------

test> use vehicles
switched to db vehicles
vehicles> show dbs
EmployeeDB      40.00 KiB
admin           40.00 KiB
animal          80.00 KiB
config         108.00 KiB
coursedb        56.00 KiB
coursedetails   40.00 KiB
demo            40.00 KiB
employeesdb     56.00 KiB
local          112.00 KiB
vehicles>  db.createCollection("two_wheelers", { capped: true, size: 4194304, max: 2000 })
{ ok: 1 }
vehicles> db.createCollection("four_wheelers")
{ ok: 1 }
vehicles> show collections
four_wheelers
two_wheelers
vehicles>
```

4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

db.two_wheelers.insert({bike_name: "Honda Shine", model: "gear", category: "125cc",colors_available: ["red", "black", "blue"], manufacturer: "Honda", performance: 7, timestamp: new Date("2020-01-01"), price: 75000})

db.two_wheelers.insert({bike_name: "TVS Jupiter", model: "gearless", category: "110cc",colors_available: ["white", "blue", "black"], manufacturer: "TVS", performance: 8, timestamp: new Date("2021-05-10"), price: 70000})

db.two_wheelers.insert({bike_name: "Yamaha FZ", model: "gear", category: "150cc",colors_available: ["black", "blue", "matte red"], manufacturer: "Yamaha", performance: 9, timestamp: new Date("2019-03-15"), price: 95000})

db.two_wheelers.insert({bike_name: "Bajaj Pulsar", model: "gear", category: "200cc",colors_available: ["black", "red", "silver"], manufacturer: "Bajaj", performance: 8, timestamp: new Date("2022-07-20"), price: 120000})

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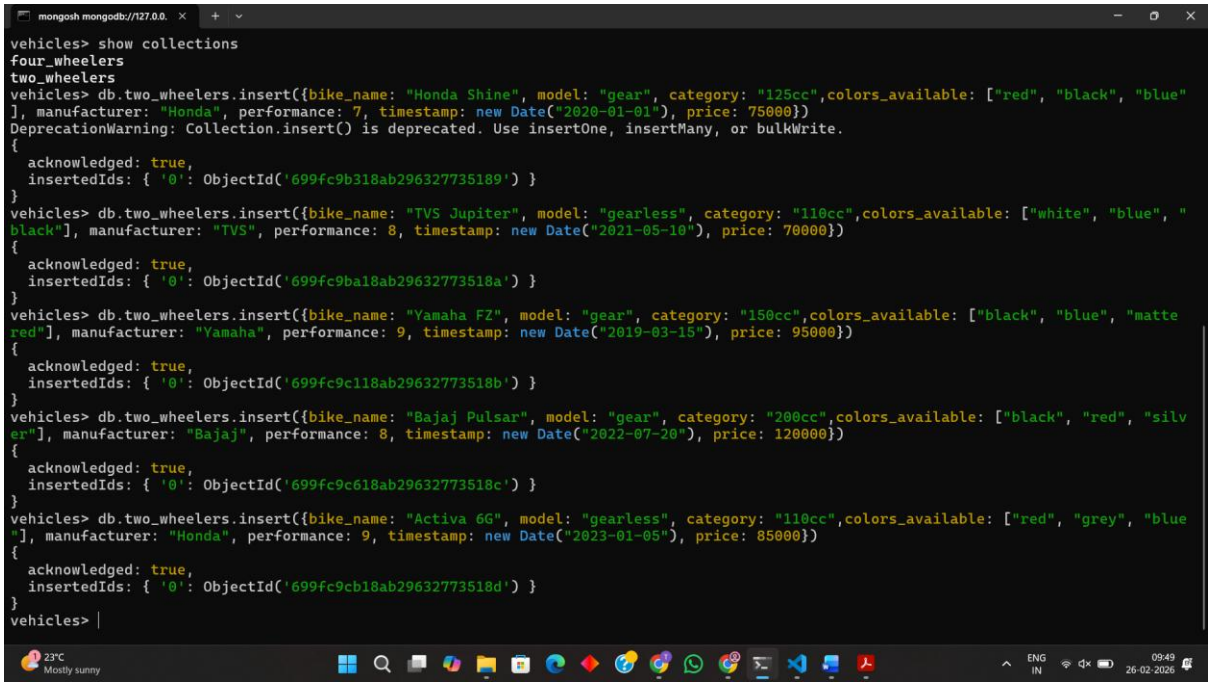
---

db.two_wheelers.insert({bike_name: "Activa 6G", model: "gearless", category: "110cc",colors_available: ["red", "grey", "blue"], manufacturer: "Honda", performance: 9, timestamp: new Date("2023-01-05"), price: 85000})



5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

db.four_wheelers.insert({vehicle_name: "Maruti Swift", model: "own", category: "car", variants: ["VXI", "ZXI", "petrol"],manufacturer: "Maruti", performance: 8, timestamp: new Date("2020-02-11"), price: 600000})

db.four_wheelers.insert({vehicle_name: "Tata Ace", model: "commercial", category: "mini truck", variants: ["diesel"], manufacturer: "Tata", performance: 7, timestamp: new Date("2019-06-25"), price: 500000})

db.four_wheelers.insert({vehicle_name: "Ashok Leyland Lorry", model: "commercial", category: "lorry", variants: ["diesel"], manufacturer: "Ashok Leyland", performance: 9, timestamp: new Date("2021-08-18"), price: 2000000})

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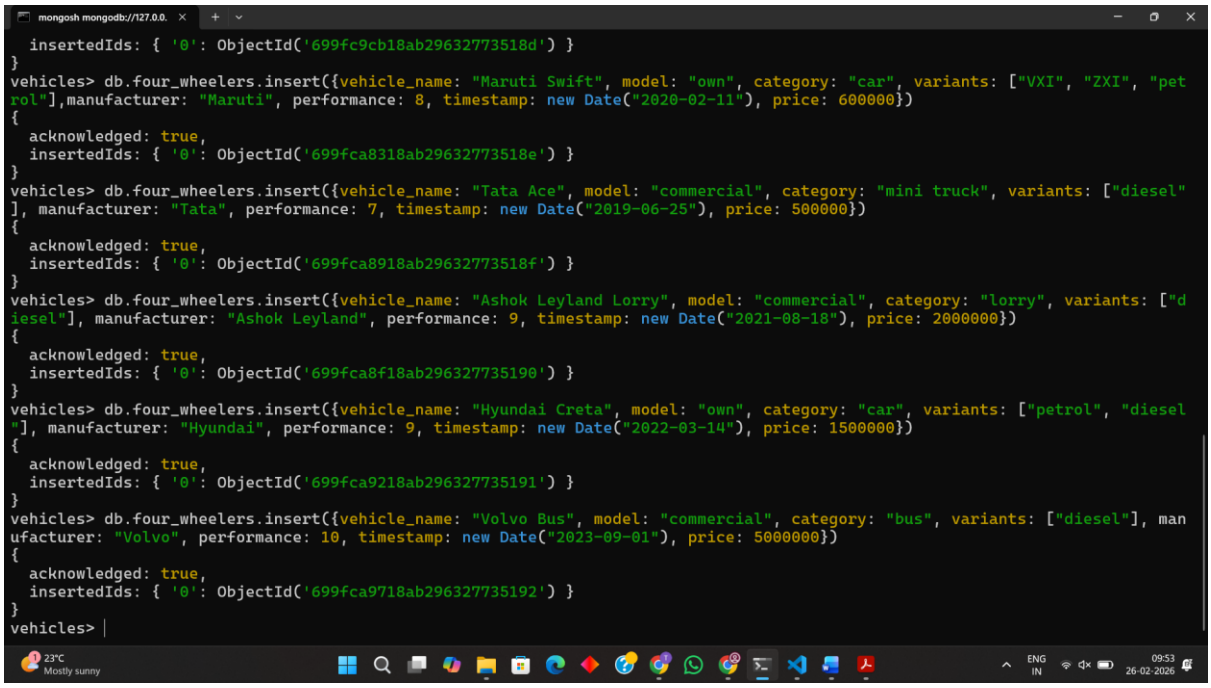
---

db.four_wheelers.insert({vehicle_name: "Hyundai Creta", model: "own", category: "car", variants: ["petrol", "diesel"], manufacturer: "Hyundai", performance: 9, timestamp: new Date("2022-03-14"), price: 1500000})

db.four_wheelers.insert({vehicle_name: "Volvo Bus", model: "commercial", category: "bus", variants: ["diesel"], manufacturer: "Volvo", performance: 10, timestamp: new Date("2023-09-01"), price: 5000000})



```
  insertedIds: { '0': ObjectId('699fc9cb18ab29632773518d') }
}
vehicles> db.four_wheelers.insert({vehicle_name: "Maruti Swift", model: "own", category: "car", variants: ["VXI", "ZXI", "pet
rol"],manufacturer: "Maruti", performance: 8, timestamp: new Date("2020-02-11"), price: 600000})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699fca8318ab29632773518e') }
}
vehicles> db.four_wheelers.insert({vehicle_name: "Tata Ace", model: "commercial", category: "mini truck", variants: ["diesel"
], manufacturer: "Tata", performance: 7, timestamp: new Date("2019-06-25"), price: 500000})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699fca8918ab29632773518f') }
}
vehicles> db.four_wheelers.insert({vehicle_name: "Ashok Leyland Lorry", model: "commercial", category: "lorry", variants: ["d
iesel"], manufacturer: "Ashok Leyland", performance: 9, timestamp: new Date("2021-08-18"), price: 2000000})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699fca8f18ab296327735190') }
}
vehicles> db.four_wheelers.insert({vehicle_name: "Hyundai Creta", model: "own", category: "car", variants: ["petrol", "diesel
"], manufacturer: "Hyundai", performance: 9, timestamp: new Date("2022-03-14"), price: 1500000})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699fca9218ab296327735191') }
}
vehicles> db.four_wheelers.insert({vehicle_name: "Volvo Bus", model: "commercial", category: "bus", variants: ["diesel"], man
ufacturer: "Volvo", performance: 10, timestamp: new Date("2023-09-01"), price: 5000000})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699fca9718ab296327735192') }
}
vehicles>
```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

db.two_wheelers.find()
db.four_wheelers.find()

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```
mongosh mongodb://127.0.0.
vehicles> db.two_wheelers.find();
[
  {
    _id: ObjectId('699fc9b318ab296327735189'),
    bike_name: 'Honda Shine',
    model: 'gear',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 7,
    timestamp: ISODate('2020-01-01T00:00:00.000Z'),
    price: 75000
  },
  {
    _id: ObjectId('699fc9ba18ab29632773518a'),
    bike_name: 'TVS Jupiter',
    model: 'gearless',
    category: '110cc',
    colors_available: [ 'white', 'blue', 'black' ],
    manufacturer: 'TVS',
    performance: 8,
    timestamp: ISODate('2021-05-10T00:00:00.000Z'),
    price: 70000
  },
  {
    _id: ObjectId('699fc9c118ab29632773518b'),
    bike_name: 'Yamaha FZ',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'black', 'blue', 'matte red' ],
    manufacturer: 'Yamaha',
    performance: 9,
    timestamp: ISODate('2019-03-15T00:00:00.000Z'),
    price: 95000
  },
  {
    _id: ObjectId('699fc9c618ab29632773518c'),
    bike_name: 'Bajaj Pulsar',
    model: 'gear',
    category: '200cc',
    colors_available: [ 'black', 'red', 'silver' ],
    manufacturer: 'Bajaj',
    performance: 8,
    timestamp: ISODate('2022-07-20T00:00:00.000Z'),
    price: 120000
  },
  {
    _id: ObjectId('699fc9cb18ab29632773518d'),
    bike_name: 'Activa 6G',
    model: 'gearless',
    category: '110cc',
    colors_available: [ 'red', 'grey', 'blue' ],
    manufacturer: 'Honda',
    performance: 9,
    timestamp: ISODate('2023-01-05T00:00:00.000Z'),
    price: 85000
```

```
mongosh mongodb://127.0.0.
vehicles> db.four_wheelers.find()
[
  {
    _id: ObjectId('699fca8318ab29632773518e'),
    vehicle_name: 'Maruti Swift',
    model: 'own',
    category: 'car',
    variants: [ 'VXI', 'ZXI', 'petrol' ],
    manufacturer: 'Maruti',
    performance: 8,
    timestamp: ISODate('2020-02-11T00:00:00.000Z'),
    price: 600000
  },
  {
    _id: ObjectId('699fca8918ab29632773518f'),
    vehicle_name: 'Tata Ace',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 7,
    timestamp: ISODate('2019-06-25T00:00:00.000Z'),
    price: 500000
  },
  {
    _id: ObjectId('699fca8f18ab296327735190'),
    vehicle_name: 'Ashok Leyland Lorry',
    model: 'commercial',
    category: 'lorry',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 9,
    timestamp: ISODate('2021-08-18T00:00:00.000Z'),
    price: 2000000
  },
  {
    _id: ObjectId('699fca9218ab296327735191'),
    vehicle_name: 'Hyundai Creta',
    model: 'own',
    category: 'car',
    variants: [ 'petrol', 'diesel' ],
    manufacturer: 'Hyundai',
    performance: 9,
    timestamp: ISODate('2022-03-14T00:00:00.000Z'),
    price: 1500000
  },
  {
    _id: ObjectId('699fca9718ab296327735192'),
    vehicle_name: 'Volvo Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Volvo',
    performance: 10,
    timestamp: ISODate('2023-09-01T00:00:00.000Z'),
    price: 5000000
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

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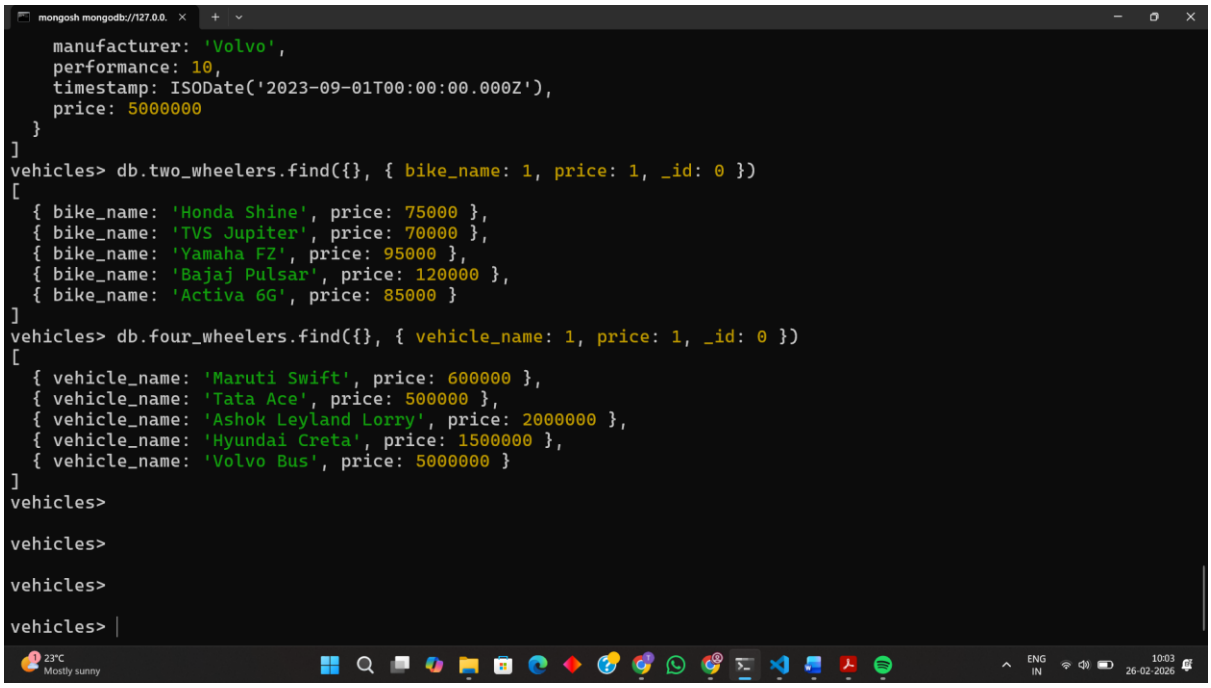
---

db.two_wheelers.find({}, { bike_name: 1, price: 1, _id: 0 })
db.four_wheelers.find({}, { vehicle_name: 1, price: 1, _id: 0 })



```
    manufacturer: 'Volvo',
    performance: 10,
    timestamp: ISODate('2023-09-01T00:00:00.000Z'),
    price: 5000000
  }
]
vehicles> db.two_wheelers.find({}, { bike_name: 1, price: 1, _id: 0 })
[
  { bike_name: 'Honda Shine', price: 75000 },
  { bike_name: 'TVS Jupiter', price: 70000 },
  { bike_name: 'Yamaha FZ', price: 95000 },
  { bike_name: 'Bajaj Pulsar', price: 120000 },
  { bike_name: 'Activa 6G', price: 85000 }
]
vehicles> db.four_wheelers.find({}, { vehicle_name: 1, price: 1, _id: 0 })
[
  { vehicle_name: 'Maruti Swift', price: 600000 },
  { vehicle_name: 'Tata Ace', price: 500000 },
  { vehicle_name: 'Ashok Leyland Lorry', price: 2000000 },
  { vehicle_name: 'Hyundai Creta', price: 1500000 },
  { vehicle_name: 'Volvo Bus', price: 5000000 }
]
vehicles>

vehicles>

vehicles>

vehicles>
```

8. Write a MongoDB query to display two_wheelers from a particular company

db.two_wheelers.find({ manufacturer: "Honda" })

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```
vehicles>

vehicles> db.two_wheelers.find({ manufacturer: "Honda" })
[
  {
    _id: ObjectId('699fc9b318ab296327735189'),
    bike_name: 'Honda Shine',
    model: 'gear',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 7,
    timestamp: ISODate('2020-01-01T00:00:00.000Z'),
    price: 75000
  },
  {
    _id: ObjectId('699fc9cb18ab29632773518d'),
    bike_name: 'Activa 6G',
    model: 'gearless',
    category: '110cc',
    colors_available: [ 'red', 'grey', 'blue' ],
    manufacturer: 'Honda',
    performance: 9,
    timestamp: ISODate('2023-01-05T00:00:00.000Z'),
    price: 85000
  }
]
vehicles>
```

9. Write a MongoDB query to display four_wheelers available in diesel variants

db.four_wheelers.find({ variants: "diesel" },{})

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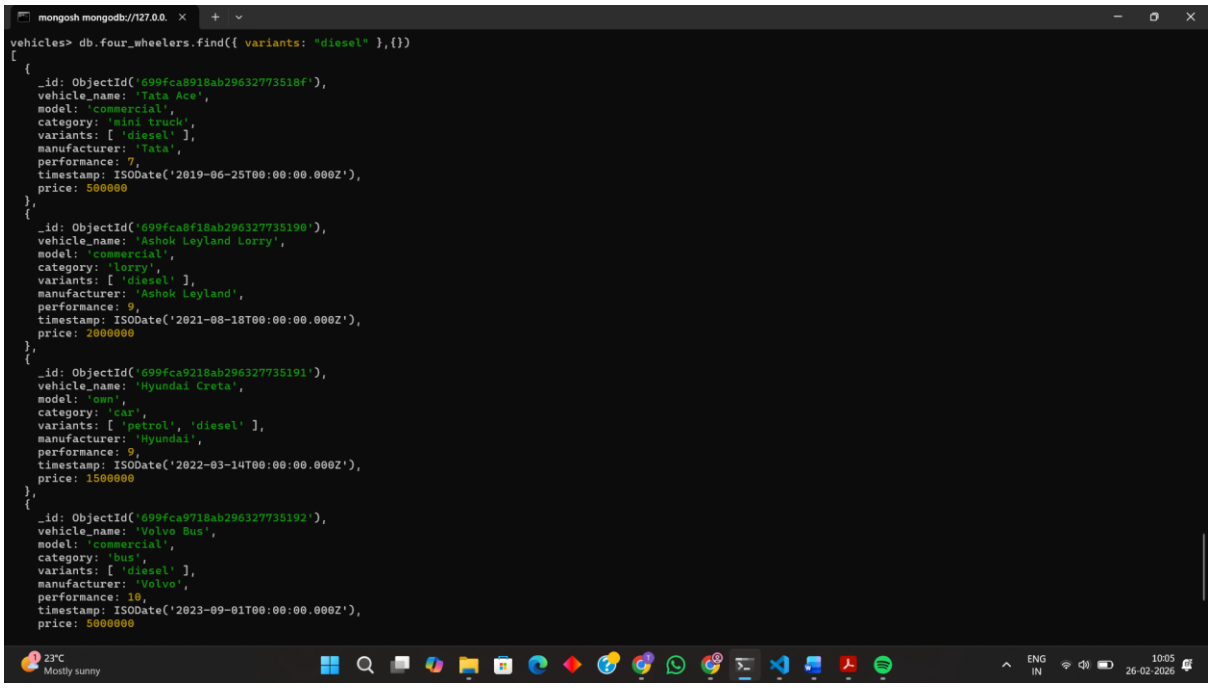

```
vehicles> db.four_wheelers.find({ variants: "diesel" },{})
[
  {
    _id: ObjectId('699fca8918ab296327735 18f'),
    vehicle_name: 'Tata Ace',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 7,
    timestamp: ISODate('2019-06-25T00:00:00.000Z'),
    price: 500000
  },
  {
    _id: ObjectId('699fca8f18ab296327735190'),
    vehicle_name: 'Ashok Leyland Lorry',
    model: 'commercial',
    category: 'lorry',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 9,
    timestamp: ISODate('2021-08-18T00:00:00.000Z'),
    price: 2000000
  },
  {
    _id: ObjectId('699fca9218ab296327735191'),
    vehicle_name: 'Hyundai Creta',
    model: 'own',
    category: 'car',
    variants: [ 'petrol', 'diesel' ],
    manufacturer: 'Hyundai',
    performance: 9,
    timestamp: ISODate('2022-03-14T00:00:00.000Z'),
    price: 1500000
  },
  {
    _id: ObjectId('699fca9718ab296327735192'),
    vehicle_name: 'Volvo Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Volvo',
    performance: 10,
    timestamp: ISODate('2023-09-01T00:00:00.000Z'),
    price: 5000000
```

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

db.two_wheelers.find({ performance: { $gt: 5 } }, { bike_name: 1, category: 1, manufacturer: 1, _id: 0 })
db.four_wheelers.find({ performance: { $gt: 5 } }, { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 })

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```
mongosh mongodb://127.0.0.

vehicles> db.two_wheelers.find( { performance: { $gt: 5 } }, { bike_name: 1, category: 1, manufacturer: 1, _id: 0 })
[
  {
    bike_name: 'Honda Shine',
    category: '125cc',
    manufacturer: 'Honda'
  },
  { bike_name: 'TVS Jupiter', category: '110cc', manufacturer: 'TVS' },
  { bike_name: 'Yamaha FZ', category: '150cc', manufacturer: 'Yamaha' },
  {
    bike_name: 'Bajaj Pulsar',
    category: '200cc',
    manufacturer: 'Bajaj'
  },
  { bike_name: 'Activa 6G', category: '110cc', manufacturer: 'Honda' }
]
vehicles> db.four_wheelers.find( { performance: { $gt: 5 } }, { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 })
[
  {
    vehicle_name: 'Maruti Swift',
    category: 'car',
    manufacturer: 'Maruti'
  },
  {
    vehicle_name: 'Tata Ace',
    category: 'mini truck',
    manufacturer: 'Tata'
  },
  {
    vehicle_name: 'Ashok Leyland Lorry',
    category: 'lorry',
    manufacturer: 'Ashok Leyland'
  },
  {
    vehicle_name: 'Hyundai Creta',
    category: 'car',
    manufacturer: 'Hyundai'
  },
  { vehicle_name: 'Volvo Bus', category: 'bus', manufacturer: 'Volvo' }
]
```

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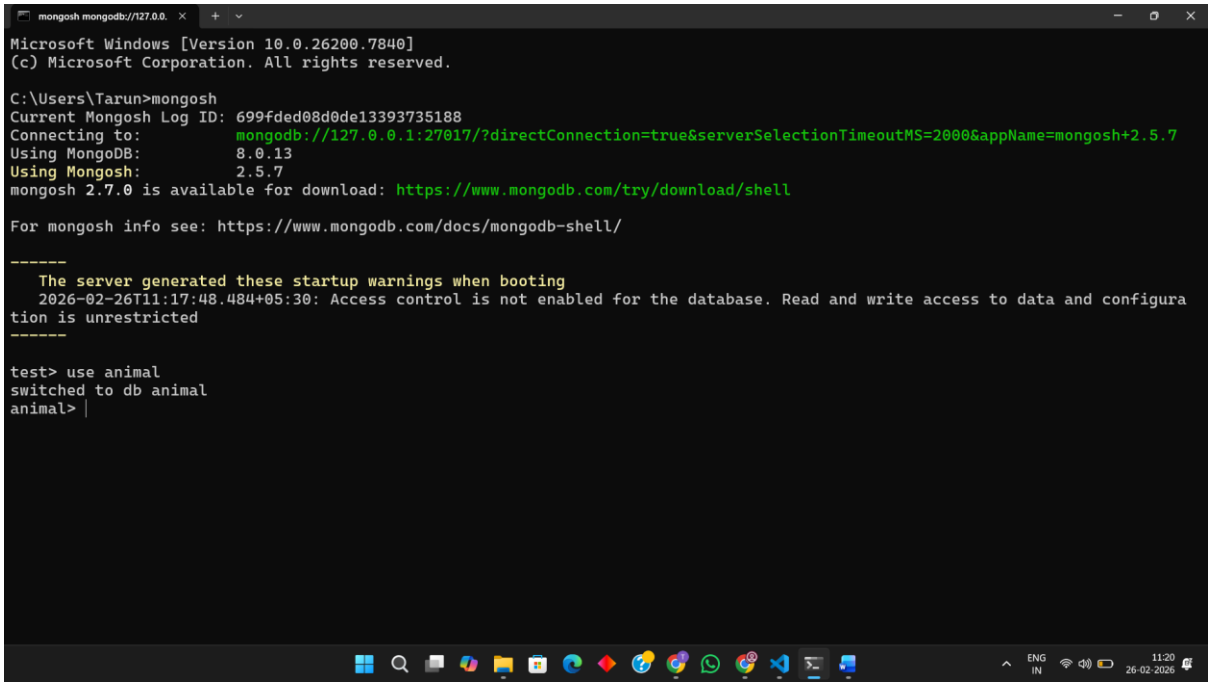
---

2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.

use animal



2. Write a MongoDB query to display all the databases.

show dbs

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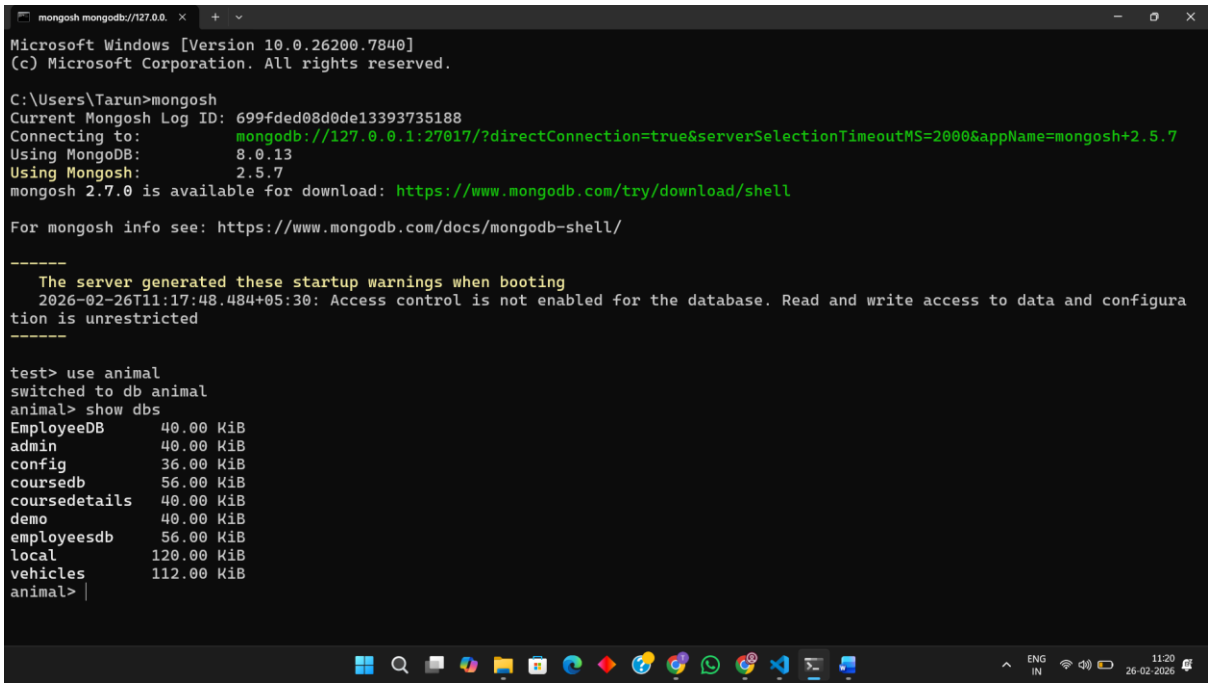

```
mongosh mongodb://127.0.0.
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Tarun>mongosh
Current Mongosh Log ID: 699fded08d0de13393735188
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.7
Using MongoDB:          8.0.13
Using Mongosh:          2.5.7
mongosh 2.7.0 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-02-26T11:17:48.484+05:30: Access control is not enabled for the database. Read and write access to data and configura
tion is unrestricted
------

test> use animal
switched to db animal
animal> show dbs
EmployeeDB      40.00 KiB
admin           40.00 KiB
config          36.00 KiB
coursedb        56.00 KiB
coursedetails   40.00 KiB
demo            40.00 KiB
employeesdb     56.00 KiB
local          120.00 KiB
vehicles       112.00 KiB
animal>
```

3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.

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Lab Sheet 6: MongoDB Basic commands

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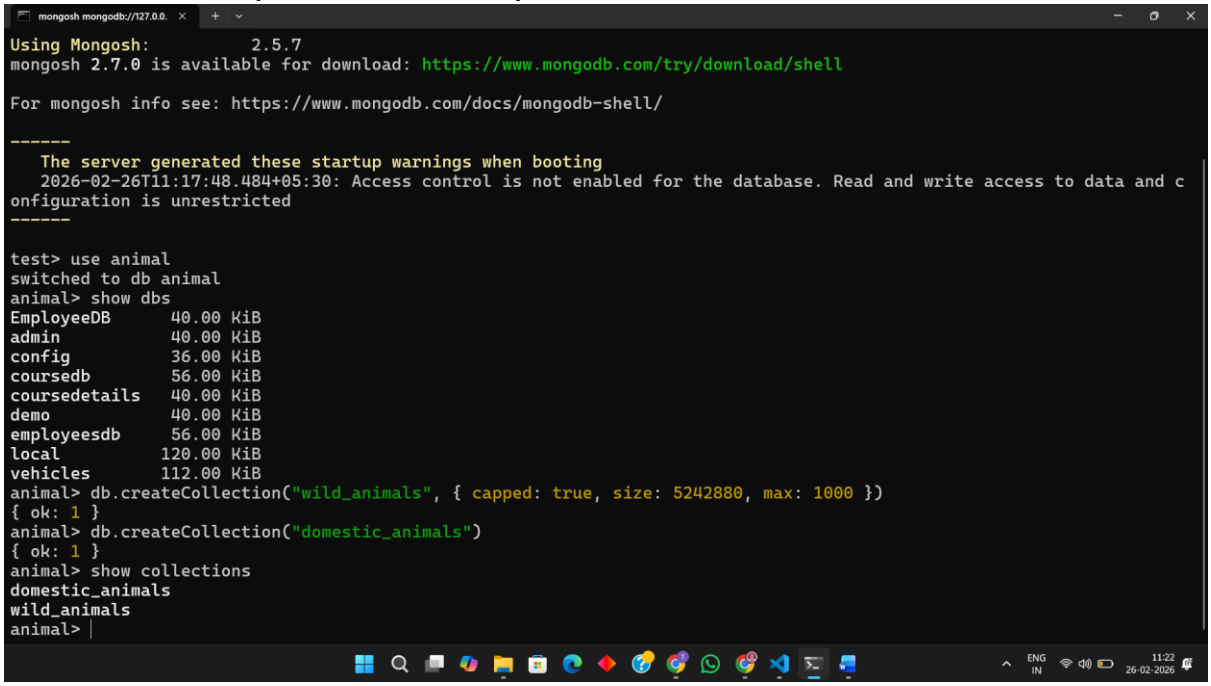
Student name: TARUN KUMAR **Reg. no.: 23BCE7563**

---

db.createCollection("wild_animals", { capped: true, size: 5242880, max: 1000 })
db.createCollection("domestic_animals")



4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

db.wild_animals.insert({animal_name: "Lion", nature: "harm", favorite_foods: ["meat", "deer"], care_taker_name: "Tarun", life_span: 12, timestamp: new Date("2021-05-15"), expenses: 50000})

db.wild_animals.insert({animal_name: "Elephant", nature: "harmless", favorite_foods: ["grass", "bananas"], care_taker_name: "Gopi", life_span: 60, timestamp: new Date("2020-08-21"), expenses: 70000})

db.wild_animals.insert({animal_name: "Tiger", nature: "harm", favorite_foods: ["meat", "rabbits"], care_taker_name: "Gopi", life_span: 15, timestamp: new Date("2022-01-10"), expenses: 65000})

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Lab Sheet 6: MongoDB Basic commands

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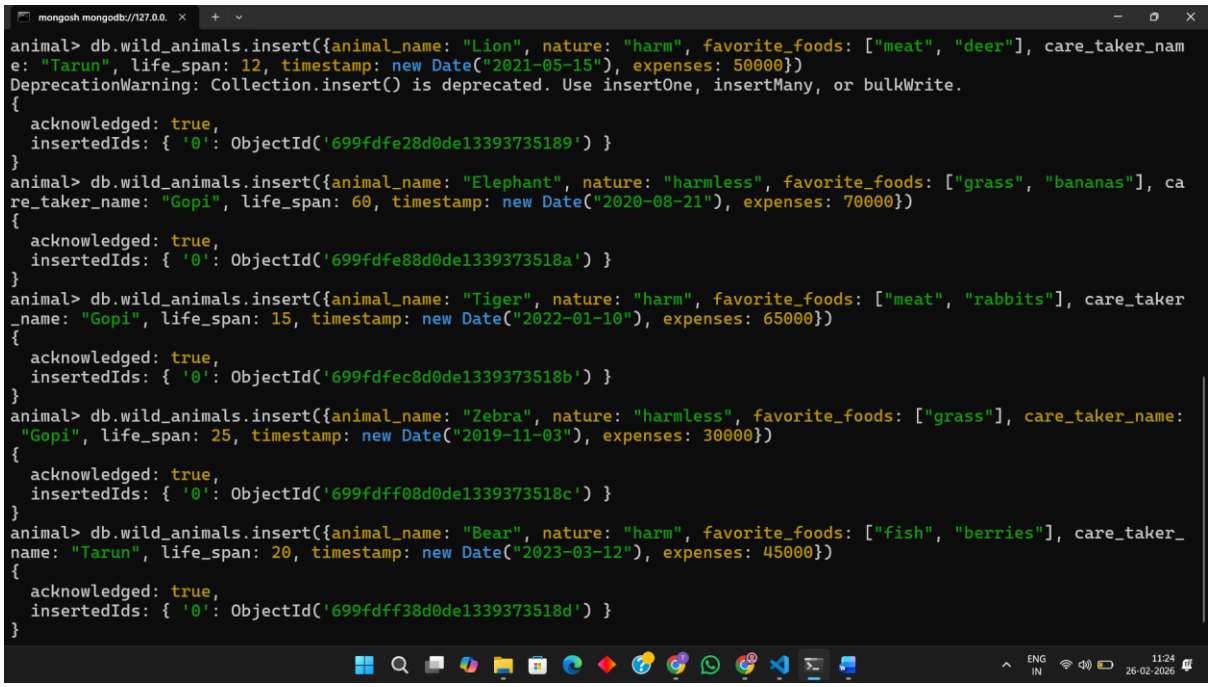
Student name: TARUN KUMAR **Reg. no.: 23BCE7563**

---

db.wild_animals.insert({animal_name: "Zebra", nature: "harmless", favorite_foods: ["grass"], care_taker_name: "Gopi", life_span: 25, timestamp: new Date("2019-11-03"), expenses: 30000})

db.wild_animals.insert({animal_name: "Bear", nature: "harm", favorite_foods: ["fish", "berries"], care_taker_name: "Tarun", life_span: 20, timestamp: new Date("2023-03-12"), expenses: 45000})



5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

db.domestic_animals.insert({animal_name: "Cow", gender: "female", favorite_foods: ["grass", "hay"], animal_petname: "Ganga", life_span: 18, timestamp: new Date("2021-02-05"), expenses: 15000})

db.domestic_animals.insert({animal_name: "Dog", gender: "male", favorite_foods: ["meat", "biscuits"], animal_petname: "Tommy", life_span: 12, timestamp: new Date("2020-07-20"), expenses: 10000})

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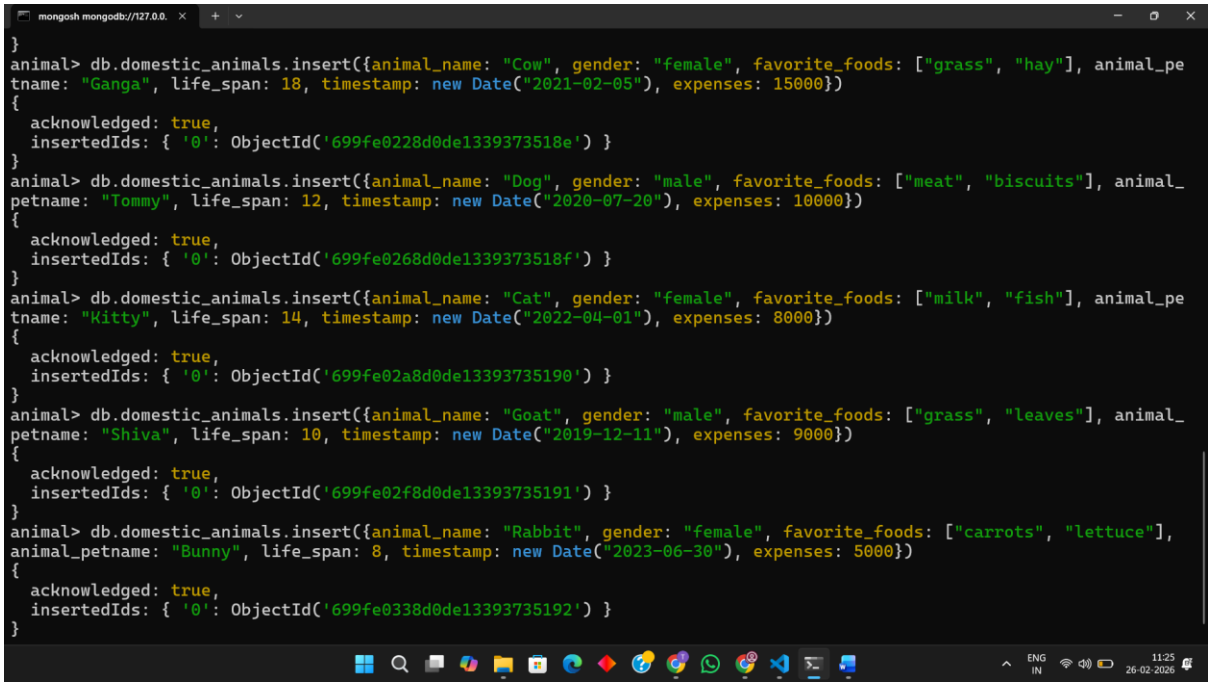
---

db.domestic_animals.insert({animal_name: "Cat", gender: "female", favorite_foods: ["milk", "fish"], animal_petname: "Kitty", life_span: 14, timestamp: new Date("2022-04-01"), expenses: 8000})

db.domestic_animals.insert({animal_name: "Goat", gender: "male", favorite_foods: ["grass", "leaves"], animal_petname: "Shiva", life_span: 10, timestamp: new Date("2019-12-11"), expenses: 9000})

db.domestic_animals.insert({animal_name: "Rabbit", gender: "female", favorite_foods: ["carrots", "lettuce"], animal_petname: "Bunny", life_span: 8, timestamp: new Date("2023-06-30"), expenses: 5000})



```
}
animal> db.domestic_animals.insert({animal_name: "Cow", gender: "female", favorite_foods: ["grass", "hay"], animal_pe
tname: "Ganga", life_span: 18, timestamp: new Date("2021-02-05"), expenses: 15000})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699fe0228d0de1339373518e') }
}
animal> db.domestic_animals.insert({animal_name: "Dog", gender: "male", favorite_foods: ["meat", "biscuits"], animal_
petname: "Tommy", life_span: 12, timestamp: new Date("2020-07-20"), expenses: 10000})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699fe0268d0de1339373518f') }
}
animal> db.domestic_animals.insert({animal_name: "Cat", gender: "female", favorite_foods: ["milk", "fish"], animal_pe
tname: "Kitty", life_span: 14, timestamp: new Date("2022-04-01"), expenses: 8000})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699fe02a8d0de13393735190') }
}
animal> db.domestic_animals.insert({animal_name: "Goat", gender: "male", favorite_foods: ["grass", "leaves"], animal_
petname: "Shiva", life_span: 10, timestamp: new Date("2019-12-11"), expenses: 9000})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699fe02f8d0de13393735191') }
}
animal> db.domestic_animals.insert({animal_name: "Rabbit", gender: "female", favorite_foods: ["carrots", "lettuce"],
animal_petname: "Bunny", life_span: 8, timestamp: new Date("2023-06-30"), expenses: 5000})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699fe0338d0de13393735192') }
}
```

6. Write a MongoDB query to display all documents available in wild_animals and domastic_animals.

db.wild_animals.find()
db.domestic_animals.find()

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Lab Sheet 6: MongoDB Basic commands

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Student name: TARUN KUMAR **Reg. no.: 23BCE7563**

```
mongosh mongodb://127.0.0.

animal> db.wild_animals.find()
[
  {
    _id: ObjectId('699fdfe28d0de13393735189'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'Tarun',
    life_span: 12,
    timestamp: ISODate('2021-05-15T00:00:00.000Z'),
    expenses: 50000
  },
  {
    _id: ObjectId('699fdfe88d0de1339373518a'),
    animal_name: 'Elephant',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'bananas' ],
    care_taker_name: 'Gopi',
    life_span: 60,
    timestamp: ISODate('2020-08-21T00:00:00.000Z'),
    expenses: 70000
  },
  {
    _id: ObjectId('699fdfec8d0de1339373518b'),
    animal_name: 'Tiger',
    nature: 'harm',
    favorite_foods: [ 'meat', 'rabbits' ],
    care_taker_name: 'Gopi',
    life_span: 15,
    timestamp: ISODate('2022-01-10T00:00:00.000Z'),
    expenses: 65000
  },
  {
    _id: ObjectId('699fdff08d0de1339373518c'),
    animal_name: 'Zebra',
    nature: 'harmless',
    favorite_foods: [ 'grass' ],
```

```
mongosh mongodb://127.0.0.

  {
    _id: ObjectId('699fdff08d0de1339373518c'),
    animal_name: 'Zebra',
    nature: 'harmless',
    favorite_foods: [ 'grass' ],
    care_taker_name: 'Gopi',
    life_span: 25,
    timestamp: ISODate('2019-11-03T00:00:00.000Z'),
    expenses: 30000
  },
  {
    _id: ObjectId('699fdff38d0de1339373518d'),
    animal_name: 'Bear',
    nature: 'harm',
    favorite_foods: [ 'fish', 'berries' ],
    care_taker_name: 'Tarun',
    life_span: 20,
    timestamp: ISODate('2023-03-12T00:00:00.000Z'),
    expenses: 45000
  }
]
animal> db.domestic_animals.find()
[
  {
    _id: ObjectId('699fe0228d0de1339373518e'),
    animal_name: 'Cow',
    gender: 'female',
    favorite_foods: [ 'grass', 'hay' ],
    animal_petname: 'Ganga',
    life_span: 18,
    timestamp: ISODate('2021-02-05T00:00:00.000Z'),
    expenses: 15000
  },
  {
    _id: ObjectId('699fe0268d0de1339373518f'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'biscuits' ],
```

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Lab Sheet 6: MongoDB Basic commands

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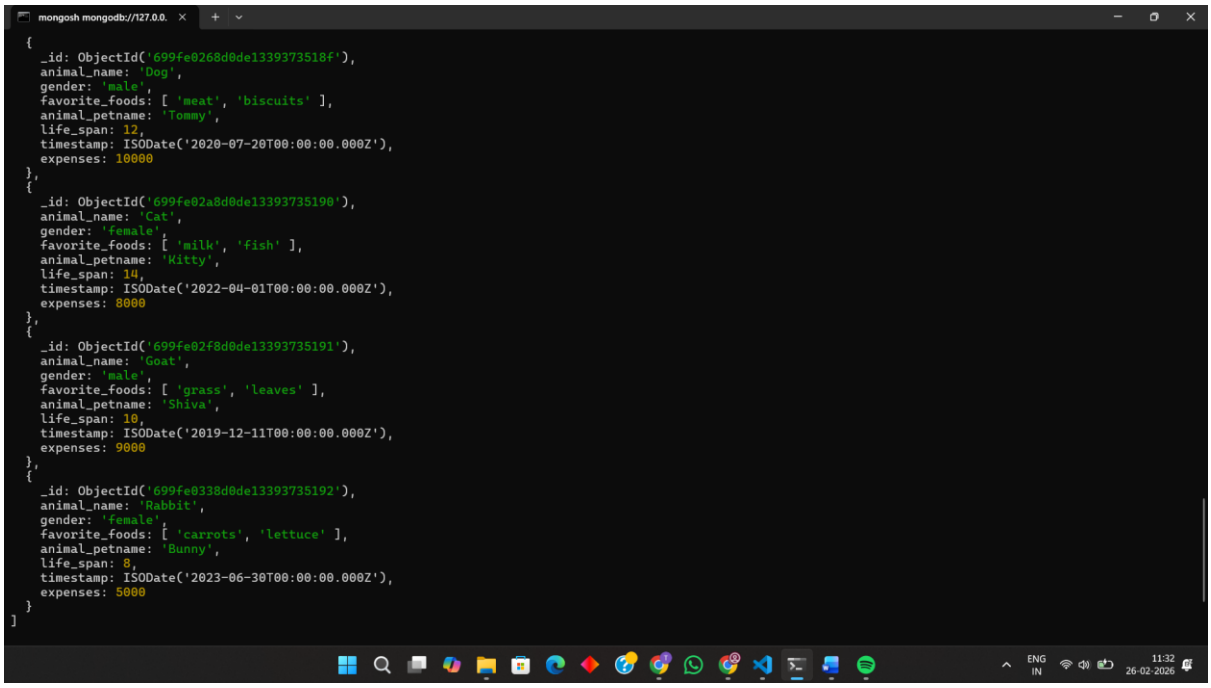
Student name: TARUN KUMAR **Reg. no.: 23BCE7563**



```
mongosh mongodb://127.0.0.
[
  {
    _id: ObjectId('699fe0268d0de1339373518f'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'biscuits' ],
    animal_petname: 'Tommy',
    life_span: 12,
    timestamp: ISODate('2020-07-20T00:00:00.000Z'),
    expenses: 10000
  },
  {
    _id: ObjectId('699fe02a8d0de13393735190'),
    animal_name: 'Cat',
    gender: 'female',
    favorite_foods: [ 'milk', 'fish' ],
    animal_petname: 'Kitty',
    life_span: 14,
    timestamp: ISODate('2022-04-01T00:00:00.000Z'),
    expenses: 8000
  },
  {
    _id: ObjectId('699fe02f8d0de13393735191'),
    animal_name: 'Goat',
    gender: 'male',
    favorite_foods: [ 'grass', 'leaves' ],
    animal_petname: 'Shiva',
    life_span: 10,
    timestamp: ISODate('2019-12-11T00:00:00.000Z'),
    expenses: 9000
  },
  {
    _id: ObjectId('699fe0338d0de13393735192'),
    animal_name: 'Rabbit',
    gender: 'female',
    favorite_foods: [ 'carrots', 'lettuce' ],
    animal_petname: 'Bunny',
    life_span: 8,
    timestamp: ISODate('2023-06-30T00:00:00.000Z'),
    expenses: 5000
  }
]
```

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

db.wild_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
db.domestic_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })

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Lab Sheet 6: MongoDB Basic commands

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Student name: TARUN KUMAR **Reg. no.: 23BCE7563**

```
  }
]
animal>

animal>

animal> db.wild_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
[
  { animal_name: 'Lion', expenses: 50000 },
  { animal_name: 'Elephant', expenses: 70000 },
  { animal_name: 'Tiger', expenses: 65000 },
  { animal_name: 'Zebra', expenses: 30000 },
  { animal_name: 'Bear', expenses: 45000 }
]
animal> db.domestic_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
[
  { animal_name: 'Cow', expenses: 15000 },
  { animal_name: 'Dog', expenses: 10000 },
  { animal_name: 'Cat', expenses: 8000 },
  { animal_name: 'Goat', expenses: 9000 },
  { animal_name: 'Rabbit', expenses: 5000 }
]
animal>
```

8. Write a MongoDB query to display domestic_animals whose life is a particular year

db.domestic_animals.find({ life_span: 8 })

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Lab Sheet 6: MongoDB Basic commands

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```
mongosh mongodb://127.0.0.
]
animal>

animal>

animal>

animal>

animal> db.domestic_animals.find({ life_span: 8 })
[
  {
    _id: ObjectId('699fe0338d0de13393735192'),
    animal_name: 'Rabbit',
    gender: 'female',
    favorite_foods: [ 'carrots', 'lettuce' ],
    animal_petname: 'Bunny',
    life_span: 8,
    timestamp: ISODate('2023-06-30T00:00:00.000Z'),
    expenses: 5000
  }
]
animal>

animal>

animal>

animal>
```

9. Write a MongoDB query to display wild_animals available under a particular care_taker

db.wild_animals.find({ care_taker_name: "Tarun" })

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Lab Sheet 6: MongoDB Basic commands

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```
mongosh mongodb://127.0.0.

animal>

animal>

animal> db.wild_animals.find({ care_taker_name: "Tarun" })
[
  {
    _id: ObjectId('699fdfe28d0de13393735189'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'Tarun',
    life_span: 12,
    timestamp: ISODate('2021-05-15T00:00:00.000Z'),
    expenses: 50000
  },
  {
    _id: ObjectId('699fdff38d0de1339373518d'),
    animal_name: 'Bear',
    nature: 'harm',
    favorite_foods: [ 'fish', 'berries' ],
    care_taker_name: 'Tarun',
    life_span: 20,
    timestamp: ISODate('2023-03-12T00:00:00.000Z'),
    expenses: 45000
  }
]
animal>
```

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

db.wild_animals.find({ life_span: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 })
db.domestic_animals.find({ life_span: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 })

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```
mongosh mongodb://127.0.0.
    expenses: 45000
  }
]
animal> db.wild_animals.find({ life_span: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 })
[
  {
    animal_name: 'Lion',
    favorite_foods: [ 'meat', 'deer' ],
    expenses: 50000
  },
  {
    animal_name: 'Elephant',
    favorite_foods: [ 'grass', 'bananas' ],
    expenses: 70000
  },
  {
    animal_name: 'Tiger',
    favorite_foods: [ 'meat', 'rabbits' ],
    expenses: 65000
  },
  {
    animal_name: 'Zebra',
    favorite_foods: [ 'grass' ],
    expenses: 30000
  },
  {
    animal_name: 'Bear',
    favorite_foods: [ 'fish', 'berries' ],
    expenses: 45000
  }
]
```

```
mongosh mongodb://127.0.0.
animal> db.domestic_animals.find( { life_span: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0
 })
[
  {
    animal_name: 'Cow',
    favorite_foods: [ 'grass', 'hay' ],
    expenses: 15000
  },
  {
    animal_name: 'Dog',
    favorite_foods: [ 'meat', 'biscuits' ],
    expenses: 10000
  },
  {
    animal_name: 'Cat',
    favorite_foods: [ 'milk', 'fish' ],
    expenses: 8000
  },
  {
    animal_name: 'Goat',
    favorite_foods: [ 'grass', 'leaves' ],
    expenses: 9000
  },
  {
    animal_name: 'Rabbit',
    favorite_foods: [ 'carrots', 'lettuce' ],
    expenses: 5000
  }
]
animal>
```